

ISOM 835 1

Predictive Analytics and Machine Learning

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Introduction & Dataset Description

This dataset from Olist is a real-world e-commerce transaction in Brazil from 2016 to 2018, which includes over 100,000 orders. It offers rich, interconnected data on orders, products, sellers, customers, reviews, payments, and delivery logistics. It captures many different parts of the e-commerce experience, from purchase behavior and payment preferences to delivery time and satisfaction.

The dataset is particularly relevant for developing practical business insights in the retail and logistics sectors. It enables exploration of common e-commerce challenges such as predicting order outcomes, identifying top-performing sellers, and analyzing customer satisfaction. Additionally, its large scale and complexity make it ideal for applying a full predictive analytics workflow.

- Orders Dataset: Contains order-level information such as order status, purchase timestamp, delivery dates, and customer IDs.
- Order Items Dataset: Provides details about each item within an order, including product IDs, seller IDs, and price.
- Order Payments Dataset: Includes payment types, payment amounts, and installment details.
- Order Reviews Dataset: Contains customer review scores (1–5 stars) and comments.
- Products Dataset: Offers metadata about products, including categories and physical dimensions.
- Customers Dataset: Provides customer locations (city, state, and zip code).
- Sellers Dataset: Contains seller locations and identifiers.

- Geolocation Dataset: Maps zip codes to geographical coordinates, used primarily for mapping.
- Product Category Translation: Translates Portuguese product category names to English.

EDA

Key Steps & Visuals:

- **Order Status Distribution**

A bar chart was used to display the distribution of order statuses. The majority of orders were marked as “delivered,” with smaller but significant counts for “canceled” and “pending” orders. These variations support the feasibility of classification modeling.

- **Delivery Time Analysis**

Delivery time was calculated as the number of days between the purchase date and the delivery date. A histogram showed that most orders were delivered within 20 days, although some outliers with longer delivery periods were observed.

- **Product Category Frequency**

A bar chart of the top 10 product categories revealed that “bed bath table” and “health beauty” were among the most frequently ordered. This provides insight into consumer preferences on the platform.

- **Review Score Distribution**

Review scores ranged from 1 to 5, with the majority clustered around 4 and 5 stars.

However, a notable number of low scores highlighted the potential for exploring drivers of customer dissatisfaction.

- **Payment Method Breakdown**

The majority of customers used credit cards, followed by boleto, vouchers, and debit cards. These insights may help identify which payment methods are linked to higher completion or satisfaction rates.

Observations:

- Some missing values were identified in timestamp fields and delivery dates.
- The dataset's richness across categories enables multi-dimensional analysis, such as linking review scores to product types or delivery times.
- The structure, which includes one-to-many relationships between customers and orders or orders and items, supports more advanced modeling and aggregation techniques.

Preprocessing

1. **Handling Missing Values:** Some timestamp fields had null values, particularly in canceled or unshipped orders. These were either filled or dropped depending on the business context.
 - Rationale: A missing delivery date for a "delivered" order indicates bad data. These rows were removed to ensure accuracy in modeling delivery performance.
2. **Feature Engineering: Delivery Time & Delay** To predict order outcomes or customer satisfaction, we engineered a new feature: delivery time in days.
 - Rationale: Delivery time is a meaningful continuous variable that can impact both customer satisfaction and order status.

3. Categorical Encoding: Several columns such as payment method, product category, and order status are categorical. Label encoding or one-hot encoding was applied depending on modeling needs
 - Rationale: Label encoding simplifies categorical variables for models like logistic regression. One-hot encoding could be used for tree-based models.
4. Merging Tables for Modeling To enrich the feature set, relevant tables were merged based on order_id.
 - Rationale: Combining customer reviews, payment data, and delivery performance offers a more holistic dataset for classification or regression models.
5. Train/Test Split and Scaling: The final dataset was split and scaled for modeling.
 - Rationale: Scaling ensures better performance for models sensitive to feature magnitude.

Business Question

1. Which product categories tend to have the longest delivery times?
 - Understanding which product categories are associated with longer delivery durations can help logistics and inventory teams optimize fulfillment processes and improve delivery reliability across key product types.
2. Do certain sellers consistently receive higher review scores than others?
 - Analyzing seller performance through review scores enables marketplace managers to identify top-performing vendors, address service inconsistencies, and refine partnership strategies.

3. Is there a relationship between the number of items in an order and customer satisfaction?
 - Larger orders may introduce more complexity in packaging, coordination, and delivery, potentially leading to longer wait times or increased risk of fulfillment issues. This can negatively affect the customer experience and result in lower satisfaction ratings.
 - By examining whether the number of items in an order correlates with review scores, user experience teams can better understand customer expectations around bulk purchases. These insights can support improvements in order handling policies or post-purchase communication strategies to enhance satisfaction across order sizes.

Modeling

1. Which product categories tend to have the longest delivery times?
 - Modeling Goal: Predict delivery time
 - Models Used: Linear Regression, Random Forest Regressor (for non-linear relationships)
 - Target Variable: `delivery_days`
 - Features: `product_category_encoded`, `num_items`
2. Do certain sellers consistently receive higher review scores than others?
 - Modeling Goal: Predict review score
 - Models Used: Decision Tree Regressor or Classifier, Random Forest Regressor or Classifier

- Target Variable: review_score
 - Features: seller_id, delivery_days, num_items
3. Is there a relationship between the number of items in an order and customer satisfaction?
- Modeling Goal: Predict review score based on order size
 - Models Used: Linear Regression or Logistic Regression, Random Forest Classifier
 - Target Variable: review_score (or review_score >= 4 as binary)
 - Features: num_items, delivery_days, maybe product_category_encoded

Performance Table:

	Business Question	Model	Metric	Score
0	Predict Delivery Days	Linear Regression	RMSE	9.060754
1	Predict Delivery Days	Random Forest Regressor	RMSE	8.951701
2	Predict Review Score by Seller	Linear Regression	RMSE	1.263263
3	Predict Review Score by Seller	Random Forest Regressor	RMSE	1.340847
4	Classify High Satisfaction	Logistic Regression	Accuracy	0.791937
5	Classify High Satisfaction	Random Forest Classifier	Accuracy	0.792256

Insights and Answers:

1. Which product categories tend to have the longest delivery times?

Key Insights:

- Some product categories (e.g., furniture, home appliances) consistently exhibited longer delivery times, likely due to size, fragility, or complex fulfillment logistics.
- Larger orders (with more items) also tended to extend delivery duration, suggesting added complexity in packaging and shipment coordination.

Implications:

- The logistics team can prioritize inventory placement for slow-shipping categories by stocking them closer to high-demand regions.
- Marketing and operations teams could adjust estimated delivery dates shown at checkout based on category and order size, improving transparency and reducing post-purchase dissatisfaction.

Limitations:

- The model does not account for distance between seller and buyer or regional warehouse efficiency, which could further refine delivery time estimates.

2. Do certain sellers consistently receive higher review scores than others?**Key Insights:**

- Even after accounting for delivery time and item count, certain sellers still stood out as consistently high or low performers in terms of review scores.
- This suggests that factors beyond logistics—such as product condition, customer service, and packaging quality—likely contribute to customer satisfaction.

Implications:

- The marketplace team can develop a seller reputation scoring system using this model's insights.

- High-performing sellers could be prioritized in search rankings or promoted through partnerships, while underperforming sellers might benefit from targeted support or quality improvement programs.

Limitations:

- The dataset lacks qualitative feedback (e.g., text reviews or return rates), so the model can't explain why certain sellers perform better—it only detects patterns in ratings.

3. Is there a relationship between the number of items in an order and customer satisfaction?

Key Insights:

- Delivery time had a stronger negative impact on satisfaction than order size—customers were more tolerant of larger orders, but delays led to noticeably lower satisfaction.
- Surprisingly, larger orders did not correlate with significantly worse experiences, countering assumptions that they create more room for error.

Implications:

- Customer service efforts should focus more on managing expectations around delivery times than on the complexity of the order itself.
- Automated flags could be set up for orders expected to exceed certain delivery thresholds, prompting proactive engagement (e.g., status updates, discounts, or apologies).

Limitations:

- Review score is a limited proxy for satisfaction—it does not capture nuances in customer sentiment.

- External factors like promotions or customer demographics were not included and could influence perceptions.

Ethics

From a business perspective, using predictive models responsibly is essential to maintaining fairness, transparency, and brand trust. For example, models that assess seller performance or predict customer satisfaction could influence high-stakes decisions such as ranking sellers on a marketplace or triggering automated service interventions. If these models are applied without transparency, they may damage relationships with sellers or mislead internal teams about the root causes of customer dissatisfaction.

Additionally, while complex models like Random Forests may offer better performance, they can be difficult to explain to business stakeholders. It's important that insights drawn from such models are accompanied by clear summaries, visuals, and business reasoning. Decision-makers need to understand not just what the model predicts, but why it matters and how it aligns with strategic goals. To ensure effective and ethical use, predictive tools should be treated as decision support, not decision replacement, and always paired with sound business judgment.

Appendix

Code

https://colab.research.google.com/drive/191HbVepT19hbMKlyNsBZqxvAU_1bJvxE?usp=sharing

Plots





